

HEALTHCONNECT APPOINTMENT ANALYTICS REPORT

Week 5 — Exploratory Data Analysis, KPI Development & Business Insights

1. Introduction

This report presents an exploratory analysis of the HealthConnect Appointment Dataset. The purpose of the analysis was to investigate appointment attendance and no-show patterns, examine relationships between appointment outcomes and relevant patient and operational characteristics, develop meaningful Key Performance Indicators (KPIs), and identify business insights that can support improved appointment management.

The analysis was conducted using Microsoft Power BI. The dataset contains information relating to patients, appointments, booking behaviour, previous appointment history, reminders, distance to the clinic, waiting time, and appointment outcomes.

The analysis focused particularly on identifying factors associated with attended, no-show, and cancelled appointments, with the objective of helping HealthConnect understand missed appointments and identify opportunities to improve attendance.

2. Data Preparation

Data preparation was conducted before beginning the exploratory analysis. The dataset was reviewed using Power Query and the available data dictionary to understand the structure, variables, data types, and potential data-quality issues.

The original dataset was not overwritten during the preparation process.

2.1 Data Quality Findings

Data Quality Area	Finding
Data types	Data types were reviewed and found to be appropriate for analysis

Appointment ID	No duplicate Appointment IDs were identified
Distance to Clinic	90 missing values were identified
Gender	Female, Male, and Prefer not to say were consistently represented
Age Group	Age groups were consistently represented
Appointment Type	Appointment categories were consistently represented
Numeric variables	No obvious invalid values were identified

Missing values in Distance to Clinic and Waiting Time were retained rather than being replaced with zero because a missing value does not necessarily indicate zero distance or zero waiting time.

Repeated Patient IDs were not treated as duplicates because the same patient may legitimately have multiple appointments.

2.2 Data Preparation Summary

Overall, the dataset was considered suitable for exploratory analysis. The main data-quality issues identified were missing values in Distance to Clinic and Waiting Time. No duplicate Appointment IDs or major categorical inconsistencies were identified.

3. Exploratory Data Analysis

Exploratory Data Analysis was conducted to investigate appointment attendance and no-show patterns across appointment characteristics, patient characteristics, previous appointment history, reminders, waiting time, distance to the clinic, cancellations, and no-show outcomes.

3.1 Appointment Outcomes

The appointment outcome analysis showed:

- No Show: 48.46%

- Attended: 46.28%
- Cancelled: 5.26%

No-shows represented the largest appointment outcome, accounting for almost half of all appointments.

This indicates that missed appointments are a significant operational issue for HealthConnect and may result in unused appointment capacity and reduced efficiency.

3.2 Appointment Characteristics

Appointment characteristics were examined through appointment type and booking lead time.

The booking lead-time analysis showed that no-show appointments were the most prominent outcome in the observed pattern. This suggests that the amount of time between booking and the scheduled appointment may be relevant when investigating attendance behaviour.

Appointment type was also examined to identify differences in appointment outcomes between different service categories.

3.3 Patient Characteristics

Patient characteristics were analysed using age group and gender.

The age-group analysis showed that the 65+ age group recorded the highest number of no-show appointments.

The gender analysis showed that female patients recorded the highest number of no-show appointments.

These findings identify demographic groups that could be investigated further when developing targeted attendance-improvement strategies.

However, the results represent appointment counts rather than necessarily showing that these groups have the highest no-show rates. Differences in the number of appointments across groups should therefore be considered.

3.4 Previous Appointment History

Previous appointment behaviour was investigated using the number of previous no-shows.

The analysis showed a decreasing pattern in no-show outcomes as previous no-shows increased in the observed data.

This finding should be interpreted cautiously because the visual primarily represents appointment counts. A rate-based analysis would provide a stronger assessment of whether previous no-show behaviour is associated with future attendance.

3.5 Reminder Analysis

Reminder behaviour was examined using reminder status and reminder channel.

The analysis showed that appointments where reminders were sent had a higher number of no-show appointments than appointments where reminders were not sent.

This does not mean that reminders cause patients to miss appointments. One possible explanation is that reminders may be targeted toward patients who are already considered more likely to miss their appointments.

Reminder channels were also compared. SMS recorded the highest number of no-show appointments, while email recorded the lowest number among the observed reminder channels.

HealthConnect could therefore investigate whether reminder timing, message content, or channel selection can be improved.

3.6 Waiting Time

Waiting time was analysed to investigate whether operational waiting periods were associated with appointment outcomes.

The analysis identified 23 minutes as the highest observed waiting-time value associated with the observed outcome pattern.

There were also 60 missing Waiting Time values in the dataset. These were not treated as zero because doing so could distort the analysis.

Further analysis using waiting-time categories could provide a clearer understanding of the relationship between waiting time and attendance.

3.7 Distance to Clinic

Distance to clinic was analysed to investigate whether physical distance may be related to appointment attendance.

The highest observed distance in the analysis was 5.80 km.

There were 90 missing Distance to Clinic values, meaning that conclusions involving distance should be interpreted using the records where distance information was available.

The analysis does not establish that distance causes patients to miss appointments.

3.8 Cancellations

Cancelled appointments represented 5.26% of all appointments.

Although cancellations were substantially lower than no-shows, they remain operationally important because cancelled appointment slots may remain unused if another patient cannot be scheduled.

4. KPI Development

Five KPIs were selected to provide a concise view of appointment performance and operational efficiency.

4.1 Total Appointments

Definition: The total number of appointment records in the dataset.

Calculation: Count of Appointment ID.

Business importance: This KPI measures the overall volume of appointments handled by HealthConnect and provides context for the other performance indicators.

4.2 No-Show Rate

Definition: The percentage of appointments where the patient did not attend.

Calculation:

No-Show Rate = No-Show Appointments ÷ Total Appointments × 100

Result: 48.46%

Business importance: This is a critical KPI because no-shows represent almost half of all appointments and may significantly affect clinic capacity and operational efficiency.

4.3 Non-Attendance Rate

Definition: The percentage of appointments that did not result in attendance, including both no-shows and cancellations.

Calculation:

Non-Attendance Rate = (No-Shows + Cancellations) ÷ Total Appointments × 100

Result: 53.72%

Business importance: This KPI provides a broader view of appointment losses by including both missed and cancelled appointments.

4.4 Cancellation Rate

Definition: The percentage of appointments that were cancelled.

Calculation:

$\text{Cancellation Rate} = \text{Cancelled Appointments} \div \text{Total Appointments} \times 100$

Result: 5.26%

Business importance: Monitoring cancellations can help HealthConnect understand how much appointment capacity is lost before scheduled consultations take place.

4.5 Average Waiting Time

Definition: The average recorded waiting time for appointments.

Calculation: Average of Waiting Time in minutes.

Business importance: Waiting time is an important operational indicator because excessive waiting may affect patient experience and overall service efficiency.

5. Dashboard Development

An interactive Power BI dashboard was developed to provide a consolidated view of appointment performance.

The dashboard contains five KPI cards at the top:

1. Total Appointments
2. No-Show Rate
3. Non-Attendance Rate
4. Cancellation Rate
5. Average Waiting Time

The dashboard also contains analytical visualisations covering:

- Appointment Outcomes Distribution
- Appointment Outcomes by Age Group
- Appointment Outcomes by Gender

- Appointment Outcomes by Reminder Status
- Appointment Outcomes by Reminder Channel
- Appointment Outcomes by Previous No-Shows
- Booking Lead Time and Appointment Outcomes
- Waiting Time and Appointment Outcomes
- Distance to Clinic and Appointment Outcomes

The dashboard allows decision-makers to examine appointment performance from multiple perspectives and identify patterns that may require further investigation.

6. Key Business Insights

The analysis produced several important findings.

6.1 No-shows are the primary challenge

No-shows accounted for 48.46% of appointments, making them the largest appointment outcome. Reducing missed appointments should therefore be a major operational priority.

6.2 More than half of appointments were non-attended

The combined non-attendance rate was 53.72%, including both no-shows and cancellations. This highlights a substantial gap between scheduled appointments and completed attendances.

6.3 Older patients showed a high number of no-shows

The 65+ age group recorded the highest number of no-show appointments. This may justify further investigation into whether older patients face specific barriers to attending appointments.

6.4 Reminder patterns require further investigation

Appointments with reminders showed a higher number of no-shows in the observed data. This should not be interpreted as evidence that reminders are ineffective because reminders may have been targeted toward higher-risk patients.

6.5 SMS recorded the highest number of no-shows

SMS had the highest number of no-show appointments among the observed reminder channels, while email had the lowest. HealthConnect could investigate whether differences are related to patient preferences, timing, message content, or the types of patients receiving each channel.

6.6 Operational factors should be monitored

Waiting time and distance to clinic were analysed as potential operational factors. The findings provide useful starting points for further analysis but do not establish causal relationships.

7. Recommendations

Based on the findings, the following recommendations are proposed.

7.1 Develop a targeted no-show reduction strategy

Because no-shows account for 48.46% of appointments, HealthConnect should make reducing missed appointments a key operational objective.

The clinic could identify higher-risk appointments and provide additional confirmation or follow-up before the appointment date.

7.2 Improve reminder strategies

HealthConnect should evaluate the effectiveness of its reminder system rather than simply measuring whether reminders were sent.

The organisation should test:

- reminder timing;
- reminder frequency;
- message content;
- SMS versus email;
- patient preferences;

- confirmation links or responses.

A controlled comparison could determine which approach produces the best attendance rate.

7.3 Investigate the 65+ age group

Since the 65+ age group recorded the highest number of no-shows, HealthConnect should investigate possible barriers affecting this group.

Potential areas for investigation include transportation, appointment communication, caregiver availability, appointment scheduling, and accessibility.

7.4 Monitor booking lead time

HealthConnect should monitor whether appointments booked further in advance have different attendance outcomes.

Where appropriate, shorter booking windows or confirmation processes could be considered for appointments with higher no-show risk.

7.5 Monitor waiting time

Waiting time should be monitored regularly as part of operational performance management.

HealthConnect could establish an acceptable waiting-time target and investigate appointments that consistently exceed the target.

7.6 Improve data completeness

The dataset contains missing values for both Distance to Clinic and Waiting Time.

HealthConnect should improve data collection procedures so that these fields are consistently captured. Better data quality would allow more reliable analysis of operational factors.

7.7 Introduce regular KPI monitoring

The five KPIs developed in this analysis should be monitored regularly through an interactive dashboard.

Tracking the KPIs over time would allow HealthConnect to determine whether interventions are actually reducing no-shows and improving attendance.

8. Limitations

Several limitations should be considered when interpreting the analysis.

8.1 Missing data

The dataset contains **90 missing Distance to Clinic values** and **60 missing Waiting Time values**. These missing records limit the completeness of analyses involving these variables.

8.2 Counts versus rates

Several visualisations were based on appointment counts. A group with a larger number of appointments may naturally have more no-shows.

Therefore, the finding that female patients or the 65+ age group recorded the highest number of no-shows does not necessarily mean they had the highest **no-show rate**.

8.3 No causal conclusions

The analysis identifies relationships and patterns but does not establish causation.

For example, the fact that appointments with reminders had more no-shows does not mean reminders caused patients to miss appointments. Reminders may have been targeted toward patients already considered high risk.

8.4 Limited operational variables

Although the dataset includes several useful variables, other factors that may influence attendance may not be captured, such as transportation availability, socioeconomic circumstances, health severity, weather, patient satisfaction, or accessibility requirements.

8.5 Historical analysis

The findings describe patterns within the available dataset and may not necessarily represent future appointment behaviour. Continued monitoring is required to determine whether the patterns remain consistent over time.

9. Conclusion

The HealthConnect appointment analysis identified no-shows as the most significant appointment management challenge, accounting for 48.46% of all appointments. The overall non-attendance rate, including cancellations, was 53.72%.

The EDA identified differences in appointment outcomes across age groups, gender, previous appointment behaviour, reminder status, reminder channel, booking lead time, waiting time, and distance to the clinic.

The analysis provides HealthConnect with a foundation for improving appointment management through targeted reminder strategies, monitoring high-risk appointment groups, improving data quality, and regularly tracking key performance indicators.

The Power BI dashboard consolidates these findings into an interactive analytical tool that can support evidence-based decision-making and ongoing performance monitoring.

Overall, the analysis demonstrates that reducing no-shows, improving reminder strategies, strengthening data collection, and monitoring appointment KPIs should be key priorities for improving HealthConnect's operational efficiency and appointment attendance.